

Math 300 Quiz 6 Spring 2024

Key

1. (8 pts) Use Cauchy's definition to show $\lim_{x \rightarrow 1} (3x^2 - 7) = -4$

Let $\varepsilon > 0$ be any positive real number

Begin by assuming $|x-1| \leq 1$ so $0 \leq x \leq 2$.

In this case $|f(x) - L| = |3x^2 - 7 + 4| = |3x^2 - 3| = |3(x+1)(x-1)| = |3x+3| \cdot |x-1|$.

The max value of $|3x+3|$ when $0 \leq x \leq 2$ will be 9.

So if $|x-1| \leq 1$ we know $|f(x) - L| \leq 9 \cdot |x-1| < \varepsilon$ if $|x-1| < \frac{\varepsilon}{9}$

Choose $\delta = \min \left\{ 1, \frac{\varepsilon}{9} \right\}$.

Therefore if $0 < |x-1| < \delta$ we have $|x-1| < 1$ and $|x-1| < \frac{\varepsilon}{9}$.

This implies $|f(x) - L| = |3x+3| \cdot |x-1| < 9 \cdot \frac{\varepsilon}{9} < \varepsilon$

Thus $\lim_{x \rightarrow 1} (3x^2 - 7) = -4$.

2. (4 pts) Use the definition for the derivative of $f(x)$ to show that $D_x[x^2 + 3] = 2x$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3 - x^2 - 3}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} = \lim_{h \rightarrow 0} 2x + h = 2x \end{aligned}$$

Over $\rightarrow$

3. (4 pts) Given $\lim_{x \rightarrow 3} f(x) = 5$ and $\lim_{x \rightarrow 3} g(x) = 2$.

$$\lim_{x \rightarrow 3} [2f(x) + \sqrt{g(x)} + 10] = \lim_{x \rightarrow 3} 2f(x) + \lim_{x \rightarrow 3} \sqrt{g(x)} + \lim_{x \rightarrow 3} 10$$

$$= 2 \cdot \lim_{x \rightarrow 3} f(x) + \sqrt{\lim_{x \rightarrow 3} g(x)} + 10 = 2 \cdot 5 + \sqrt{2} + 10$$

$$= 20 + \sqrt{2}$$

4. (4 pts) Given $f(2) = 7$, and $f'(2) = -5$ what is $h'(2)$ if $h(x) = \frac{f(x)}{x}$?

$$h'(x) = \frac{x \cdot f'(x) - f(x)}{x^2}$$

$$h'(2) = \frac{2 \cdot (-5) - 7}{4} = \frac{-17}{4}$$